Cap-Local Descendant Budgets and Spectral Tail Certificates for Half-Plane Double Dimers

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M426

October 9, 2026

Abstract

The source's half-plane double-dimer argument separates finite transfer algebra, a signed planar color identity, localization, and identification of loop curves. We preserve that separation and improve two finite interfaces. A recursive weight on the forest of eligible cap arches controls the total weight of all proper descendants. It replaces the source's depth weight with a uniformly larger guaranteed minimum, while remaining measurable from one uncolored cap alone. Combined with the source's color identity and transfer normalization, it gives a one-interval violation bound with cutoff factor $CM^4 4^M$, rather than the source's superexponential dependence obtained from $(2M)^{-(M-1)}$. A second result truncates the propagated excitation states with an explicit remainder retaining the interval factor $(|S|/n)^2$. We also extract a conditional Gaussian traversal tail and a near-linear short-interval modulus after the lattice-mesh upper limit. These estimates do not identify the continuum loop law. Classical exterior algebra and forest recursion are credited as tools; positivity, boundary caps, probability normalization, and the omitted geometric steps are stated explicitly. Contribution assessment: C3/5, the cap-local descendant construction and interval-preserving spectral certificate yield a notable source-specific quantitative refinement without independently proving curve convergence or CLE identification.

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1 The model, measure, and proof boundaries

The source [1] takes two independent uniform perfect matchings on corner-rooted Temperleyan square-lattice rectangles, passes to the specified half-plane infinite-volume law, and removes doubled edges. The remaining loops are unrooted and unoriented for the final curve topology. For signed calculations, however, the matching labels fix an orientation: first-matching edges go from black to white, second-matching edges from white to black. All intersection indices use the same transverse orientation.

Our finite rectangle is

$$\{(\delta x, \delta y) : 1 \le x \le m,\ 1 \le y \le n\} \setminus \{(\delta, \delta)\}, \qquad n = 2p + 1,$$

with m, n odd. Fixed physical windows and positive buffers are held constant as $\delta \downarrow 0$. A transfer length counts columns, not physical distance. The source's compactification is $F(z) = (z-i)/(z+i)$, and its loop topology matches every loop above each positive disk diameter, including multiplicity and traversal order. Our estimates below concern finite rectangles and their mesh upper limits; this topology is not replaced by a finite collection of puncture observations.

1.1 A complete source-text map

All nine cached public text files were freshly read: 2,301 literal lines, including the scientific proofs, wrapper, references, figure source, and README. The figure's TeX was read; no claim of viewing its compiled source image is made.

File and lines	Mechanism and remaining boundary
`10-introduction`, 1–259	Half-plane law, nested curve topology, cylindrical convergence input, and the distinction between a signature and a curve.
`20-transfer`, 1–518	Alternating one-particle transfer, exterior sectors, odd zero mode, deleted-corner cap normalization, marked vectors and crossing recursion.
`30-short-segment`, 1–546	Signed pair insertion, same/different-loop color ratios, cap-local weights, pointwise detection and ordered union-bound limits.
`40-localization`, 1–494	Wilson/Temperley coupling, oriented local path preservation, multiscale time changes and preliminary compactness.
`50-identification`, 1–218	Fixed puncture records, conditional coupling, two slow-grid selections, exclusion of invisible paths and backtracking, interior tightness.

Reading these proofs is not independent verification of their entire chain or of all cited literature. In particular our probability corollaries explicitly retain the source's cap/color identities and crossing recursion. Localization, cylindrical convergence, and the full curve-identification argument are not consequences of our finite operator certificate.

2 The actual transfer and its physical norm

Let $H_n = \mathbb{R}^n$ have its counting-measure Euclidean inner product. In the increasing-index orthonormal wedge basis, the column matrix is

$$A_{ij} = \mathbf{1}_{\{i=j\}} + \mathbf{1}_{\{j \text{ odd},\ |i-j|=1\}}. \tag{1}$$

Successive columns use $A, A^{\mathsf{T}}, A, \ldots$, and the j-particle transfer is $\Lambda^j A$ or its transpose. The initial and final p-particle sets are $\alpha = \{3, 5, \ldots, 2p+1\}$ and $\beta = \{2, 4, \ldots, 2p\}$. The missing corner is encoded by the virtual incoming edge, not by a periodic boundary condition. In particular

$$Z = \langle e_\beta, T^{(p)}_{m,0} e_\alpha \rangle, \qquad Z_{\mathrm{dd}} = Z^2.$$

The virtual edge occurs in both replicas and supplies no open single strand.

2.1 Why singular values, not eigenvalues of one step

Split H_n into odd and even rows. The hop block has singular values $\lambda_l = 2\cos(\pi l/(2p+2))$, $1 \le l \le p$, and one odd zero mode. Indeed its even-row square has diagonal 2 and neighboring entries 1; substituting the normalized sine vectors diagonalizes this tridiagonal matrix. In each odd/even pair,

$$A_l = \begin{pmatrix} 1 & 0 \\ \lambda_l & 1 \end{pmatrix}, \qquad u_l = \operatorname{arsinh}\left(\cos\frac{\pi l}{n+1}\right).$$

Its singular values are e^{u_l}, e^{-u_l}, whereas its eigenvalues are both one. Thus the eigenvalues of a single triangular step would miss the propagation cost completely. If $A = U\Sigma V^{\mathsf{T}}$, the first step sends V coordinates to U coordinates and the next transpose sends them back. Both have the same positive diagonal Σ in these compatible bases. The zero singular mode has singular value one, and must not be dropped.

Write $s = \exp(\sum_l u_l)$ and index the positive energies from weakest to strongest by

$$\epsilon_r = u_{p+1-r} = \operatorname{arsinh}\left(\sin\frac{\pi r}{n+1}\right), \qquad 1 \le r \le p.$$

The elementary inequalities $\sin x \ge 2x/\pi$ on $[0, \pi/2]$ and $\operatorname{arsinh} x \ge x/\sqrt{2}$ on $[0, 1]$ give

$$\epsilon_r \ge \frac{\sqrt{2}\, r}{n+1} \ge \frac{r}{\sqrt{2}\, n}. \tag{2}$$

This merely makes an existing source gap constant explicit. For a one-replica occupied mode set J let

$$w_t(J) = s^{-t} \prod_{j \in J} \sigma_j^t.$$

Relative to the filled expanding sea, a hole or a contracting particle of index r costs $e^{-t\epsilon_r}$. The zero mode is freely occupied or empty. Consequently

$$W_t := \sum_J w_t(J) = 2 \prod_{r=1}^{p} (1 + e^{-t\epsilon_r})^2. \tag{3}$$

All individual weights are at most one. For $t \ge \eta n$,

$$W_t \le 2\exp\left(\frac{2q}{1-q}\right), \qquad q = e^{-\eta/\sqrt{2}}.$$

The sum here includes all sectors; restricting to the sectors used by the source can only decrease it.

2.2 Caps and a declared source interface

For cap vectors $L_a = T^{(p)}_{a,0} e_\alpha$ and $R_b = (T^{(p)}_{m,b})^{\mathsf{T}} e_\beta$, retain the source bound

$$\frac{\|L_a \otimes L_a\| \|R_b \otimes R_b\| s^{2(b-a)}}{Z_{\mathrm{dd}}} \le C_{\mathrm{cap}}, \tag{4}$$

where $a, m-b \ge \eta n$ and m/n is bounded below. Its proof keeps the corner coefficient $|h_0| = (p+1)^{-1/2}$ and gives

$$Z = |h_0| \Big(\prod_l a_l^2 \Big) s^m \prod_l (1 - e^{-2(m+1)u_l}), \qquad a_l^2 = \frac{e^{u_l}}{2\cosh u_l}.$$

Other corner-hole terms are orthogonal in cap norms but cannot reach the all-even final wedge. Removing them from a cap norm would therefore be an invalid simplification.

An eligible cap arch reaches its prescribed deep line. On E_M each cap has at most M eligible arches. The source's marked-cap lemma allows *any* mark weight in $[0,1]$ depending only on its uncolored cap and chosen arch, and gives

$$\|V_L\| \le M \|L_a \otimes L_a\|, \qquad \|V_R\| \le M \|R_b \otimes R_b\|. \tag{5}$$

No conditioning on the other cap or the final loop connections is permitted when forming these vectors.

3 A forest budget for all descendants

Disjoint cap arches have laminar seam intervals: they are nested or disjoint. Regard the eligible arches as a rooted forest under strict interval containment. All descendants below refer to eligible arches, not to all unmarked pieces of the matching.

Theorem 3.1 (Cap-local descendant weights)**.** *Fix an integer cutoff $M \ge 1$ and $0 < \gamma < 1$. In a forest of at most M eligible arches define, from leaves upward,*

$$W_A = 1 + \gamma^{-1} \sum_{B \subsetneq A} W_B, \qquad q = \left(\frac{\gamma}{1+\gamma} \right)^{M-1}, \qquad w_A = q W_A. \tag{6}$$

Then $q \le w_A \le 1$, and

$$\sum_{B \subsetneq A} w_B = \gamma (w_A - q) < \gamma w_A. \tag{7}$$

The weights are determined from one cap alone. If every eligible arch of an inner separating loop g' is strictly spanned by an eligible arch of an outer loop g, then their cap totals satisfy

$$U_{g'} \le \gamma U_g. \tag{8}$$

The same assertion holds independently for the other cap totals V.

Proof. Let $b = 1 + \gamma^{-1}$. In a subtree with k nodes, list its proper descendants in postorder, so all descendants of the jth listed node precede it. Induction gives $W_{B_j} \le b^{j-1}$: the recursion is bounded by $1 + \gamma^{-1} \sum_{i<j} b^{i-1} = b^{j-1}$. Thus the root satisfies $W_A \le b^{k-1} \le b^{M-1}$. Every $W_A \ge 1$, proving the asserted range; the equality (7) is exactly the recursion.

Among the eligible arches of g choose those maximal by containment. They have disjoint descendant sets. By the hypothesis every eligible arch of g' is a proper descendant of one of these

maximal arches: extend any containing eligible g arch to a maximal one. Sum (7) over those maximal arches. It bounds the total weight of the inner arches by γ times the sum of their maximal outer weights, which is at most γU_g. This reasoning does not require the inner arches to be an antichain. □

The spanning hypothesis is the precise planar statement proved in the source's cap-weight-decrease argument: close an outer cap arch by its seam interval, use parity of its Jordan disks inside the cap, and use disjointness to trap the inner arch. Reaching the deep line then makes the containing outer arch eligible. We retain that statement explicitly rather than deducing it from a matrix norm.

The recurrence can be evaluated from subtree totals: if $S_A = W_A + \sum_{B \subsetneq A} W_B$ and C_A is the sum of S_B over immediate children, then $W_A = 1 + C_A/\gamma$ and $S_A = 1 + (1 + \gamma^{-1})C_A$. This requires a finite bottom-up pass once the forest is known; no measured runtime or global sampling speedup is claimed.

For $\gamma = 1/2$ a chain of three eligible arches, from deepest to outermost, has $W = (1, 3, 9)$ and $w = (1/9, 1/3, 1)$. Three disjoint eligible arches instead each have weight $1/9$. The uniform floor is deliberately conservative across forest shapes. The source's depth rule has floor $(2M)^{-(M-1)}$, whereas (6) has floor $3^{-(M-1)}$ at the same decay ratio. For $M = 3$ these floors are $1/36$ and $1/9$. We do not claim the new weights are optimal among all positive observables.

4 From signed colors to a quantitative detector

For an interval I on the middle seam, the source sums a signed two-particle insertion over a marked left arch and a marked right arch. Its exact uncolored color identity has the following form. On one loop the contribution before the cap weights is $uv/2$, where $u, v \in \mathbb{Z}$ are the two mark-to-mark indices in the same left-to-right orientation and $|u - v| \le 1$. Thus $uv \ge 0$. For distinct loops the contribution is $-t_g t_{g'}/4$, where $t_g \in \{0, 1, -1\}$. Nonzero t_g occur only on loops separating the endpoints of I. Those loops form at most two nested chains, with alternating signs in each chain and opposite outermost signs if both chains exist.

These are planar color facts, not consequences of an operator norm. The actual marked observable, with arbitrary admissible cap weights, is denoted $X_{M,I}$. Its different-loop part is

$$-\frac{1}{4} \sum_{g \neq g'} U_g V_{g'} t_g t_{g'}. \tag{9}$$

We reprove the required real-variable positivity with its free parameter.

Lemma 4.1 (Two chains with ratio γ)**.** *Let signs alternate along at most two finite chains, with opposite outermost signs. Suppose $U, V \ge 0$ each contract by a factor at most $\gamma \in (0, 1)$ at each step. Then (9) is nonnegative. If every positive U_g or V_g is at least q, and some distinct pair has $U_g V_{g'} > 0$, then (9) is at least*

$$\frac{(1 - \gamma)^2 q^2}{4}. \tag{10}$$

Proof. For a nonnegative sequence with successive ratio at most γ, its alternating sum starting positive lies between $(1 - \gamma)$ times its first term and that first term. Group consecutive positive-minus-negative pairs for the lower bound; group after its first term for the upper bound. Within a single chain, grouping the pairs by their outer member gives

$$\frac{1}{4} \sum_i \left[U_i \sum_{j>i} (-1)^{j-i+1} V_j + V_i \sum_{j>i} (-1)^{j-i+1} U_j \right].$$

Each summand is nonnegative. If $U_iV_j > 0$ for some $j > i$, then $V_{i+1} > 0$ and this group contributes at least $(1-\gamma)q^2/4$; the reversed product is identical.

Between the chains the expression is $-1/4$ times the sum of the two products of their signed U and V sums. Each nonzero sum has its chain's outermost sign and absolute value at least $(1-\gamma)$ times its outermost weight. The signs of the chains are opposite. A positive cross-chain product therefore contributes at least (10). The weaker of the within-chain and cross-chain bounds proves the result. □

Theorem 4.2 (Improved cap-local detection)**.** *Retain the source's uncolored color identity, spanning property, and marked-vector interpretation on E_M. Use the weights (6). Then $X_{M,I} \geq 0$ for every uncolored configuration, and*

$$X_{M,I} \geq d_{M,\gamma}\, \mathbf{1}_{E_M \cap \{\text{violation on } I\}}, \qquad d_{M,\gamma} = \frac{(1-\gamma)^2}{4}\left(\frac{\gamma}{1+\gamma}\right)^{2(M-1)}. \tag{11}$$

A violation means precisely either source event: a deep-to-deep arc has nonzero index different from its whole loop's index, or two such arcs on distinct loops both have nonzero index.

Proof. Theorem 3.1 supplies the chain hypotheses separately in both caps. Every positive cap total is at least q. Lemma 4.1 makes the entire different-loop sum nonnegative, even though individual terms can be negative. All same-loop terms are nonnegative by the integer observation above.

In the first violation, select the eligible cap arches containing the two deep endpoints. If the arc index is a and its whole-loop index is e, the same-loop contribution is $w_L w_R a(a-e)/2$. Here $a \neq 0, e$ and $e \in \{-1, 0, 1\}$; the integer $a(a-e)$ is positive. The contribution is at least $q^2/2$, which exceeds $d_{M,\gamma}$. If only the second violation occurs, both arc indices equal their whole-loop indices. The two loops then separate the interval endpoints, and each has eligible arches in both caps. A distinct positive product exists, so Lemma 4.1 applies. Set the observable to zero off E_M, as in the source. □

4.1 The improved cutoff factor

The source's normalized transfer estimate, (4), and (5) imply, uniformly over cap-local weights in $[0, 1]$,

$$|\mathbb{E}X_{M,I}| \leq CM^2(|S(I)|/n)^2.$$

Here C depends on the fixed rectangle and positive buffers, not on M, γ, δ, I. Combining this analytic estimate with *pointwise* (11) yields

$$\mathbb{P}(E_M \cap \{\text{violation on } I\}) \leq \frac{4CM^2}{(1-\gamma)^2}\left(\frac{1+\gamma}{\gamma}\right)^{2(M-1)}\left(\frac{|S(I)|}{n}\right)^2. \tag{12}$$

This inference would be invalid for an arbitrary signed insertion.

At $\gamma = 1/2$ the improved detection floor is $3^{-2(M-1)}/16$, compared with the source floor $(2M)^{-2(M-1)}/16$. For $M = 3$, the improvement is a factor 16. For $M \geq 2$, a convenient parameter is $\gamma = 1 - 1/M$:

$$\left(\frac{1+\gamma}{\gamma}\right)^{2(M-1)} = 4^{M-1}\left(1 + \frac{1}{2(M-1)}\right)^{2(M-1)} \leq e\, 4^{M-1}.$$

Thus (12) is at most

$$C'M^4 4^M (|S(I)|/n)^2. \tag{13}$$

For this particular guaranteed floor the unique optimal parameter when $M > 1$ solves

$$\gamma^2 + M\gamma - (M-1) = 0, \qquad \gamma_* = \frac{\sqrt{M^2+4M-4} - M}{2} \in (0,1).$$

This follows by differentiating $\log d_{M,\gamma}$; its derivative changes sign once. It is optimization of our bound, not of all possible observables. The simple choice above avoids an irrational parameter in exact computations.

5 A spectral certificate that keeps the interval factor

Let Q be the real orthogonal singular-mode matrix at the middle seam, the same for both replicas. For a site set $S = S(I)$ put $P_S = Q^{\mathsf{T}} \mathbf{1}_S Q$. In increasing wedge order the coefficient for deleting a mode pair R and creating a pair T is

$$\sum_{i<j,\ i,j\in S} \det Q_{\{i,j\},R} \det Q_{\{i,j\},T} = \det(P_S)_{R,T}. \tag{14}$$

This is the classical Cauchy–Binet identity. Occupation-state signs remain in the full insertion matrix $\mathcal{O}_S$; no positivity of that matrix is asserted.

Define the computable scalar

$$B_S = \max_{|R|=|T|=2} |\det(P_S)_{R,T}|.$$

If $|Q_{ir}| \le C_0/\sqrt{n}$, Gram Cauchy–Schwarz on two-fold exterior vectors gives

$$B_S \le C_0^4 (|S|/n)^2, \qquad B_S = 0 \text{ if } |S| < 2. \tag{15}$$

Indeed $|\det(P_S)_{R,T}|^2 \le \det(P_S)_{R,R} \det(P_S)_{T,T}$, and each principal minor is at most the product of its two diagonal entries. This removes the factor two of the source's entrywise minor estimate, but that minor constant change alone is not the main contribution.

For $0 \le L \le p$, retain states whose holes and contracting particles have indices at most L; retain either zero-mode occupancy. Put

$$W_{t,L} = 2\prod_{r=1}^{L} (1 + e^{-t\epsilon_r})^2, \qquad S_t = W_t^2, \qquad S_{t,L} = W_{t,L}^2.$$

The latter sums are for two replicas. The sector-restricted orthogonal projections onto these retained states are denoted Π_-, Π_+. Let $D_\pm$ be the diagonal normalized two-replica propagation matrices, with entries $w_{t_\pm}(J) w_{t_\pm}(K)$ in the relevant input/output sectors. Write $\mathcal{A} = D_+ \mathcal{O}_S D_-$ and $\mathcal{A}_L = \Pi_+ \mathcal{A} \Pi_-$.

Theorem 5.1 (Interval-preserving excitation remainder)**.** *In the physical orthonormal wedge norms,*

$$\|\mathcal{A}_L\| \le \|\mathcal{A}\| \le \|\mathcal{A}_L\| + B_S(S_{t_+} S_{t_-} - S_{t_+,L} S_{t_-,L}), \tag{16}$$

$$\|\mathcal{A} - \mathcal{A}_L\| \le 4 B_S S_{t_+} S_{t_-} \left(\frac{q_+^{L+1}}{1-q_+} + \frac{q_-^{L+1}}{1-q_-} \right), \tag{17}$$

where $t_\pm \ge \eta_\pm n > 0$ and $q_\pm = e^{-\eta_\pm/\sqrt{2}}$. At $L = p$ the exact remainder is zero.

Proof. For fixed input and output occupation states, their set differences identify the removed and added pairs uniquely. Hence the insertion entry has magnitude at most B_S, including its wedge sign. The propagated entry is bounded by $B_S d_+(\nu) d_-(\mu)$. The operator norm is at most the absolute sum of its entries. For an entry omitted by either projection, this sum is bounded by the all-sector product minus its retained product. This proves the remainder in (16); compression is contractive and the triangle inequality gives its two norm bounds.

By (2),

$$\frac{S_{t,L}}{S_t} = \prod_{r>L} (1 + e^{-t\epsilon_r})^{-4}.$$

Use $\log(1+x) \le x$ and $1 - e^{-x} \le x$ to obtain $S_t - S_{t,L} \le 4S_t \sum_{r>L} e^{-t\epsilon_r} \le 4S_t q^{L+1}/(1-q)$. Finally $ab - a_0 b_0 = (a - a_0)b + a_0(b - b_0)$ with $a_0 \le a$ proves (17). □

The core has at most 2^{4L+2} two-replica basis states before sector restrictions, independent of n for fixed L. This is still exponential in L, not a claim of efficient full enumeration. At fixed positive propagation widths the certified truncation error decays geometrically in L while retaining $(|S|/n)^2$. An additional computed core error $\varepsilon_{\text{core}}$ must be added separately; to preserve a short-interval estimate, that error also needs appropriate interval scaling. The zero mode remains present even at $L = 0$. Dropping it changes $W_{t,L}$ and is not an admissible truncation.

6 Conditional quantitative short-interval control

For completeness, the source's traversal recursion contains a stronger tail than its stated tightness consequence. We isolate exactly the input used, so that the following is not mistaken for an independent proof of its marked-arch geometry.

Lemma 6.1 (Closing the inherited recursion quantitatively)**.** *Assume the source's fixed-slab recursion and terminal seam bound, so that, for sufficiently large integers k, its mesh upper limit is at most*

$$C \sum_{j \ge 0} 2^j \exp\{A k_j \log(2k_j) - B 4^{-j} k_j^2\}, \qquad k_j = k^{2^j}, \tag{18}$$

with fixed $A, B, C > 0$. Then there are $a, C_1 > 0$ and k_0 such that this bound is at most $C_1 e^{-ak^2}$ for all $k \ge k_0$.

Proof. For $k \ge 8$, the sequence $k_j/[4^j \log(2k_j)]$ is increasing: its successive ratio satisfies

$$\frac{k_j \log(2k_j)}{4 \log(2k_j^2)} \ge \frac{k_j}{8} \ge 1.$$

Choose $k_0 \ge 8$ with $A \log(2k)/k \le B/2$ for all $k \ge k_0$. Then the positive term in each exponent is at most half its negative term. Moreover $4^{-j} k_j^2 \ge 4^j k^2$, because each successive ratio is $k_j^2/4 \ge 4$. Taking $a = B/2$, and increasing k_0 so that $ak_0^2 \ge 1$, bounds the sum by

$$C e^{-ak^2} \sum_{j \ge 0} 2^j e^{-3ak^2 j} \le \frac{C}{1 - 2e^{-3}} e^{-ak^2}.$$

Here $4^j \ge 1 + 3j$ was used. This proves the claim. □

For the actual source recursion A absorbs $2k_j \log 2 + 4k_j \log k_j$ and B is proportional to the fixed slab width divided by transverse size. It stops when $2k_j > n$, so there is no residual terminal probability. Rounding the subdivisions is justified only after fixing k and taking sufficiently small mesh. Since more than M eligible cap arches force at least $2M$ traversals in a fixed outer subslab, the retained recursion yields

$$\limsup_{\delta \downarrow 0} \mathbb{P}(E_M^c) \le C_2 e^{-aM^2} \quad (M \ge M_0). \tag{19}$$

Constants may be changed for the two caps.

Corollary 6.2 (A rectangle modulus after the mesh upper limit)**.** *Assume the source interfaces used in Theorem 4.2 and Lemma 6.1. For the source event $B_{\delta,h}$, formed from $O(h^{-1})$ intervals of length at most h, there are fixed positive constants such that, for all sufficiently small h,*

$$\limsup_{\delta \downarrow 0} \mathbb{P}(B_{\delta,h}) \le C(1 + \log(1/h))^2 \, h \exp\left(C' \sqrt{\log(1/h)} \right). \tag{20}$$

In particular the upper limit is $O_\varepsilon(h^{1-\varepsilon})$ for every fixed $\varepsilon > 0$.

Proof. At fixed $M \ge 2$, (13), $|S(I)|/n \le C(h + \delta)$, and the union bound give the explicit source-shaped inequality

$$\mathbb{P}(B_{\delta,h}) \le \mathbb{P}(E_M^c) + CM^4 4^M \frac{(h+\delta)^2}{h}. \tag{21}$$

Take $\limsup_{\delta \downarrow 0}$ with M, h fixed, and use (19). For each h, choose

$$M = \max\left(M_0, 2, \left\lceil \sqrt{\log(1/h)/a} \right\rceil \right).$$

The first term is at most $C_2 h$, and the second is bounded as in (20). The last assertion follows because $\sqrt{\log(1/h)}$ and $\log\log(1/h)$ are $o(\log(1/h))$. □

This is a quantified *iterated-limit* consequence of the source recursion and the new observable. There is no simultaneous mesh-size threshold stated for $M(h)$, and no uniform joint (δ, h) rate. The half-plane localization theorem chooses a rectangle after an error tolerance; without a quantitative localization bound in that tolerance, (20) is not a half-plane rate and still less a rate of curve convergence.

7 Worked normalization and comparison tests

For $n = m = 3$, the corner deletion leaves eight vertices. Here $p = 1$, $\alpha = \{3\}$, $\beta = \{2\}$, and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Z = (AA^{\mathsf{T}}A)_{2,3} = 4, \qquad Z_{\mathrm{dd}} = 16.$$

Direct domino enumeration gives the same count. An insertion on a single site vanishes: two distinct occupied sites cannot be removed. This exact rank fact agrees with $B_S = 0$ in (15). At this width the relevant two-replica sectors are $(2,0)$ and $(0,2)$, so the insertion is exactly the second exterior power of the interval projection, after transporting both factors to the common middle basis. Its entries can have signs after that orthogonal change; the color argument is still needed.

At $M = 3, \gamma = 1/2$, the forest detector floor is $1/1296$ whereas the source floor is $1/20736$. Their ratio is exactly 16, before any spectral or cap constants are changed. This is a comparison of rigorous guarantees, not an empirical change in the actual violation probability. For a fixed small interval, the excitation cutoff changes neither its site set nor the cap weights. Its error is proportional to the same B_S as the exact operator, making it useful for a finite certificate even as the number of transverse sites grows.

7.1 Fresh original Ram comparisons

The private physical-metrics chapter [6], freshly read in full, derives $G = J^*J$, the physical adjoint, and the orthogonal decomposition of a full residual into retained and omitted parts. Here all wedge norms initially use counting measure and orthonormal ordered wedges. In arbitrary coefficient bases $J_{\mathrm{in}}, J_{\mathrm{out}}$, a transfer must instead be measured from G_{in} to G_{out}; its adjoint is $G_{\mathrm{in}}^{-1} T^* G_{\mathrm{out}}$. Ignoring those metrics invalidates the compression estimate. The bound (16) explicitly accounts for the omitted physical operator, not merely its retained block.

The original RamCS chapter [7], also freshly read in full, derives fixed-contour inversion and weighted Plancherel under stated integrability hypotheses, keeps finite-interval boundary terms, and distinguishes a symbol representation from diagonalization. Those distinctions support the choice of compatible singular bases for alternating A, A^{T}, rather than pretending that the triangular one-column spectrum controls propagation. We do not introduce an unproved complex contour transform for the random loops. Its subsidiary input chapters and cited foundational reports were not opened in this fresh scope; the proofs used here are those literally present in the two read originals.

8 Advantages, limits, and public context

The local contribution is not the sine basis, exterior-power identity, or a generic change of coordinates. It is the cap-local descendant budget compatible with the source's signed planar observable, together with an interval-preserving omitted-mode certificate. The former changes the cutoff dependence from a power of M with exponent proportional to M to a fixed exponential times M^4. The latter makes a finite propagation calculation interpretable without losing its small-interval normalization. Both retain the deleted corner and the odd zero mode.

The costs remain substantial. A cap forest has to be known, and its weights depend on the cutoff and the complete eligible-arch nesting inside that cap. The spectral core remains exponentially large in its cutoff. Source cap constants and geometric buffers are fixed, not uniform over degenerating rectangles. The strongest probability consequences remain conditional on the stated color, spanning, and recursion interfaces. No experiment measures a speedup, and no finite fixture certifies localization, puncture convergence, or the two slow-grid identifications of traces and traversal order.

Public primary records were refreshed on October 9, 2026, using only public titles and URLs. Rasmussen–Ruelle [2] describe sector decomposition of Lieb's transfer matrix, not the full random-curve identification at issue here. Basok–Chelkak [3], in the September 10, 2020 version of their preprint and the 2021 JEMS paper, treat cylindrical-event probabilities. Bai–Wan [4] give the needed continuum crossing input for that identification with nested CLE. Basok–Izyurov [5], January 2, 2025, prove local nesting fluctuations and convergence of a centered nesting field. A distribution-valued nesting field is not a law of loop curves. Only primary abstracts and metadata of those external papers were read in this refresh; their full proofs remain imported. The source itself asserts the stronger full curve theorem; this report does not independently establish it.

Future work could optimize weights over a fixed observed forest, certify approximate mode bases in their physical Gram metrics, or derive quantitative Wilson localization constants. The first would improve a detector; the second would improve a numerical certificate; the third is required before transporting the rectangle modulus into a half-plane rate. None follows merely by extending a finite matrix.

9 Verification and assessment

The read ledger records all nine public files and their actual bounded display ranges. Fresh original-private reads cover the 210-line physical-metrics chapter and the 373-line RamCS chapter: 583 lines. Included-but-unopened subsidiary chapters and the physical-metrics figure are explicitly excluded. Pre-read hashes were persisted before these literal reads and compared with post-read hashes. The parent batch identity is captured-tree provenance, not a live-branch or mathematical verification claim.

Owned exact rational checks test forest descendant budgets, generalized two-chain positivity, finite transfer counts and exterior identities, and propagated truncation sums. Cases and individual assertions are recorded separately. These manufactured finite fixtures passed in 16,037 cases with 74,583 assertions. The transfer group independently enumerates perfect matchings of 16 corner-rooted rectangles; the spectral-certificate group uses rational orthogonal surrogate bases, not the source's trigonometric mode coordinates. These tests supplement the proofs; they are not samples from the half-plane law or tests of CLE convergence. Parent checks are not included in the author totals.

The contribution assessment is C3, independently of applicability L3: the explicit descendant construction is a notable local change to an actual probabilistic detection interface, and the residual certificate preserves a source-critical normalization. Classical algebra and the source's already established formulas are not credited as new. The retained global dependencies preclude any claim of a new independent solution of the scaling-limit conjecture.

References

[1] OpenAI, *The curve scaling limit of half-plane double dimers*, September 23, 2026. Pinned public source. Local hashes are recorded in the audit.

[2] J. Rasmussen and P. Ruelle, *Refined conformal spectra in the dimer model*, J. Stat. Mech. (2012), P10002. Primary preprint record.

[3] M. Basok and D. Chelkak, *Tau-functions à la Dubédat and probabilities of cylindrical events for double-dimers and CLE(4)*, J. Eur. Math. Soc. 23 (2021), 2787–2832. Primary preprint, version 2; journal record.

[4] T. Bai and Y. Wan, *On the crossing estimates for simple conformal loop ensembles*, Int. Math. Res. Not. (2023), 11645–11683. Primary preprint, version 3.

[5] M. Basok and K. Izyurov, *Nesting of double-dimer loops: local fluctuations and convergence to the nesting field of CLE(4)*, January 2, 2025. Primary preprint, version 1.

[6] Private Ram corpus, volume 21, original physical-metrics chapter, lines 1–210 freshly read. Confidential local proof comparison.

[7] Private Ram corpus, revised volume 1, original RamCS chapter, lines 1–373 freshly read; subsidiary input files not read. Confidential local proof comparison.